



Introduction to Azure OpenAI

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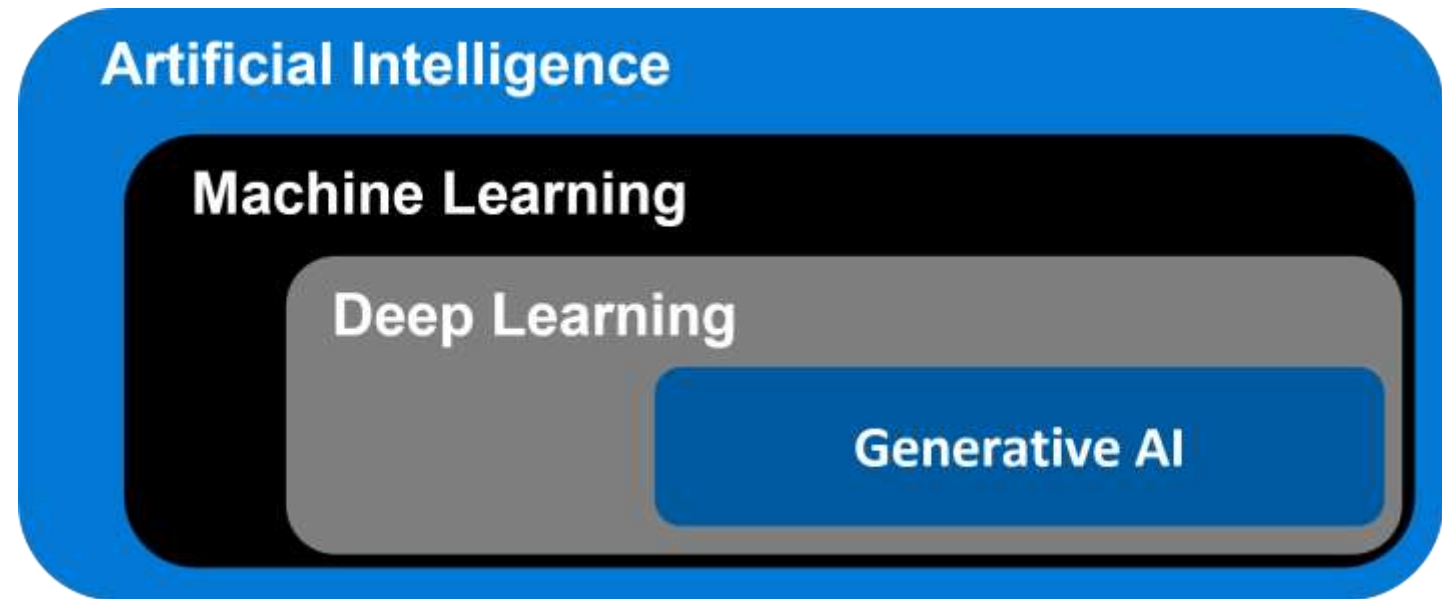
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What is generative AI

Generative AI models are a subset of deep learning models that can produce new content based on what is described in the input. The OpenAI models are a collection of generative AI models that can produce language, code, and images.



Power your apps with large-scale AI models with Azure

take advantage of large-scale, generative AI models with deep understandings of language and code to enable new reasoning and comprehension capabilities for building cutting-edge applications. Apply these coding and language models to a variety of use cases, such as writing assistance, code generation, and reasoning over data. Detect and mitigate harmful use with built-in responsible AI and access enterprise-grade Azure security.

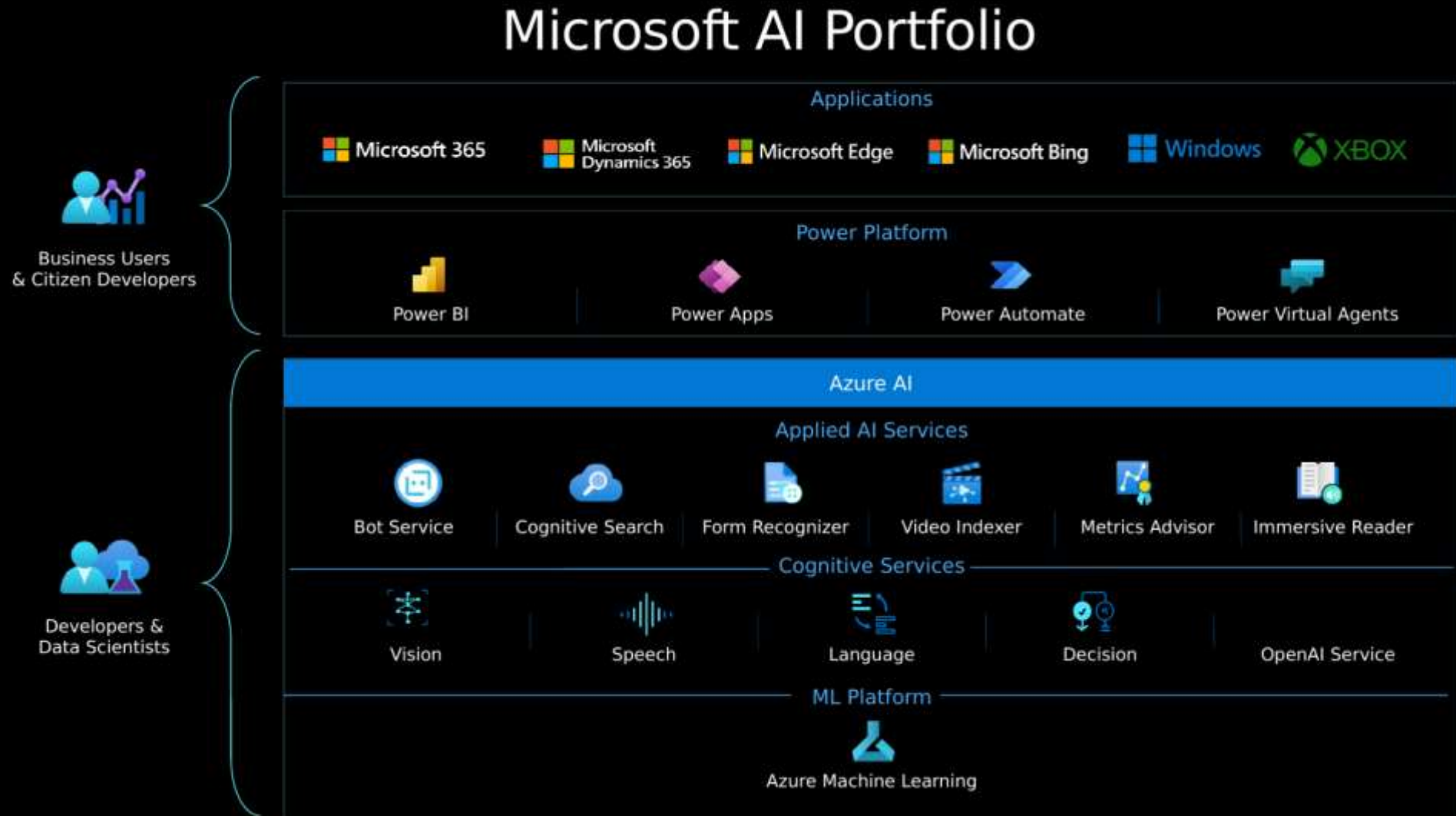
Custom AI models fine-tuned with your data and hyperparameters

Built-in features to help ensure you're using AI responsibly

Enterprise-grade security with role-based access control (RBAC) and private networks



Microsoft AI Portfolio



Azure—the best place to build AI workloads



Five of the top 50 ranked supercomputers worldwide are running on **Microsoft Azure** and **NVIDIA A100** GPUs.



Twelve of the top 20 ranked supercomputers worldwide use **NVIDIA A100** GPUs.



Introduction to Azure OpenAI Service



Azure OpenAI Service provides REST API access to OpenAI's powerful language models including the GPT-3, Codex and Embeddings model series.

Pre-trained generative AI models

Customization capabilities; the ability to fine-tune AI models with your own data

Built-in tools to detect and mitigate harmful use cases so users can implement AI responsibly

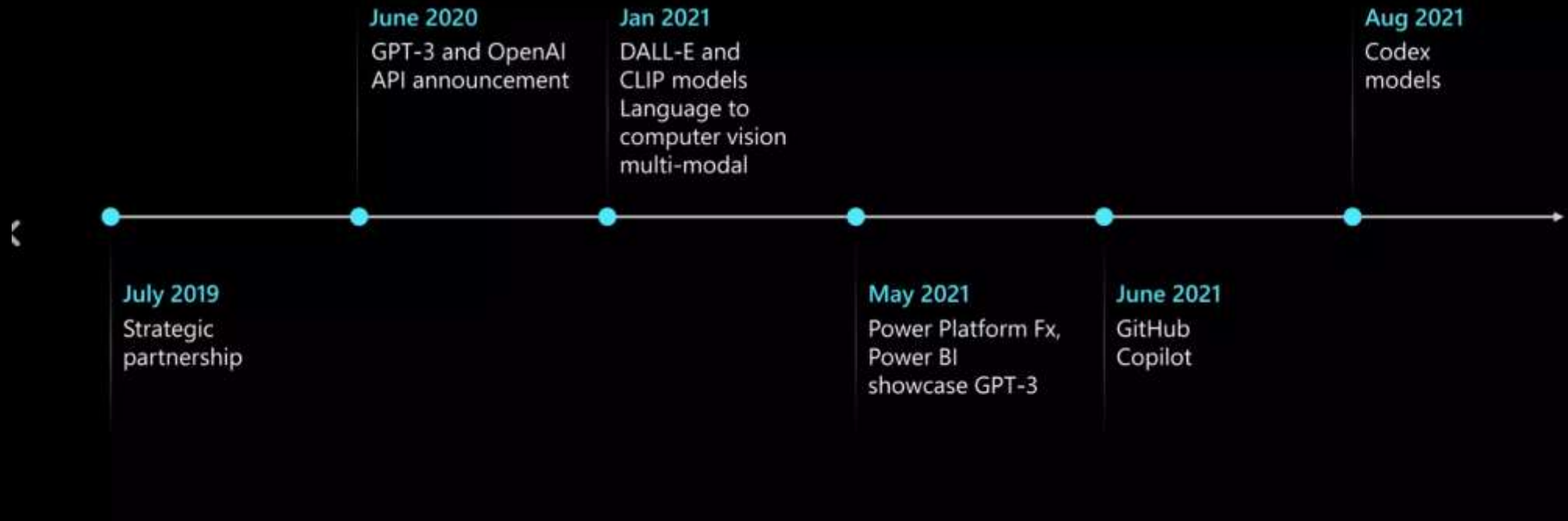
Enterprise-grade security with role-based access control (RBAC) and private networks

Workload azure Open AI Support

- Generating Natural Language
- Text completion: generate and edit text
- Embeddings: search, classify, and compare text
- Generating Code: generate, edit, and explain code
- Generating Images: generate and edit images

History

OpenAI and Microsoft



Examples AI

Q&A

Answer questions based on existing knowledge.

Grammar correction

Correct sentences into standard English.

Summarize for a 2nd grader

Translates difficult text into simpler concepts.

Natural language to OpenAI API

Create code to call the OpenAI API using a natural language instruction.

Text to command

Translate text into programmatic commands.

English to other languages

Translates English text into French, Spanish, and Japanese.

Natural language to Stripe API

Create code to call the Stripe API using natural language.

SQL translate

Translate natural language to SQL queries.

Parse unstructured data

Create tables from long-form text

Classification

Classify items into categories via example.

How to use

Cognitive Services | Azure OpenAI Studio - Preview

[Azure OpenAI Studio](#)

[Privacy & cookies](#)

Get started with Azure OpenAI

Perform a wide variety of natural language tasks with Azure OpenAI, including copywriting, summarization, parsing unstructured text, classification, and translation.

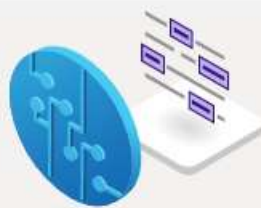
Explore examples for prompt completion



Summarize Text

Summarize text by adding a 'tl;dr:' to the end of a text passage.

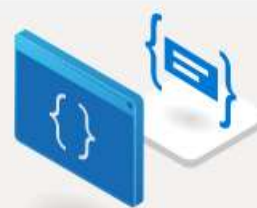
[Learn more](#)



Classify Text

Classify items into categories provided at inference time.

[Learn more](#)



Natural Language to SQL

Translate natural language to SQL queries.

[Learn more](#)



Generate New Product Names

Create product names from examples words.

[Learn more](#)

<https://aka.ms/oaiapply>

How to use

The screenshot shows the Azure OpenAI Studio GPT-3 playground interface. The top navigation bar is blue with the text "Cognitive Services | Azure OpenAI Studio" and icons for notifications, settings, help, and a user profile. The left sidebar contains a "Playground" section with "GPT-3" selected, and a "Management" section with links to "Deployments", "Models", and "File Management". The main area is titled "GPT-3 playground" and includes a "Deployments" dropdown menu set to "text-davinci-002" and an "Examples" dropdown menu set to "Load an example". A "View code" link is also present. The central text input area is labeled "Start typing here". The right sidebar, titled "Parameters", contains several settings: "Temperature" (slider from 0 to 1, set to 1), "Max length (tokens)" (slider from 0 to 100, set to 100), "Stop sequences" (text input field), "Top probabilities" (slider from 0 to 1, set to 0.5), "Frequency penalty" (slider from 0 to 1, set to 0), "Presence penalty" (slider from 0 to 1, set to 0), "Best of" (slider from 0 to 1, set to 1), "Pre-response text" (checkbox and text input field), and "Post-response text" (checkbox and text input field). A "Learn more" link is at the bottom of the parameters section. At the bottom of the main area, there are buttons for "Generate", "Undo", and "Regenerate", along with a "Tokens: 0" indicator.

<https://aka.ms/oaiapply>

How to use

GPT-3 playground

Deployments

text-davinci-002

Examples

Summarize Text

 View code

A neutron star is the collapsed core of a massive supergiant star, which had a total mass of between 10 and 25 solar masses, possibly more if the star was especially metal-rich.[1] Neutron stars are the smallest and densest stellar objects, excluding black holes and hypothetical white holes, quark stars, and strange stars.[2] Neutron stars have a radius on the order of 10 kilometres (6.2 mi) and a mass of about 1.4 solar masses.[3] They result from the supernova explosion of a massive star, combined with gravitational collapse, that compresses the core past white dwarf star density to that of atomic nuclei.

Tl;dr;

A neutron star is the collapsed core of a supergiant star. These incredibly dense objects are incredibly fascinating due to their strange properties and their potential for phenomena such as extreme gravitational forces and a strong magnetic field.

Generate

Undo

Regenerate

Tokens: 189 ⓘ

<https://aka.ms/oaiapply>

These OpenAI models fall into a few main families:



Models available

the GPT-3

A series of models that can understand and generate natural language. This includes the new ChatGPT model.

Codex

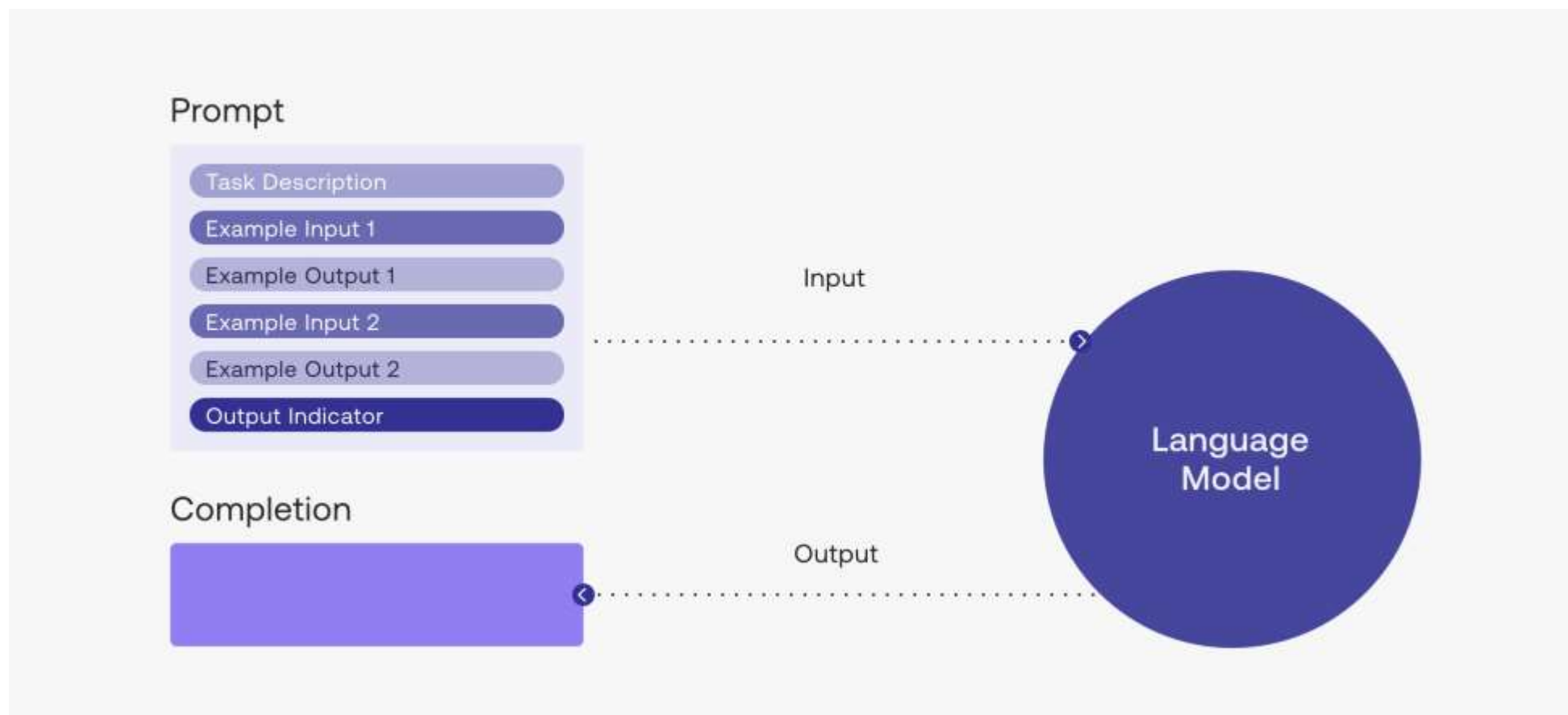
A series of models that can understand and generate code, including translating natural language to code.

Embeddings model series

A set of models that can understand and use embeddings. An embedding is a special format of data representation that can be easily utilized by machine learning models and algorithms. The embedding is an information dense representation of the semantic meaning of a piece of text. Currently, we offer three families of Embeddings models for different functionalities: similarity, text search, and code search.

Understanding prompts, completions, and tokens

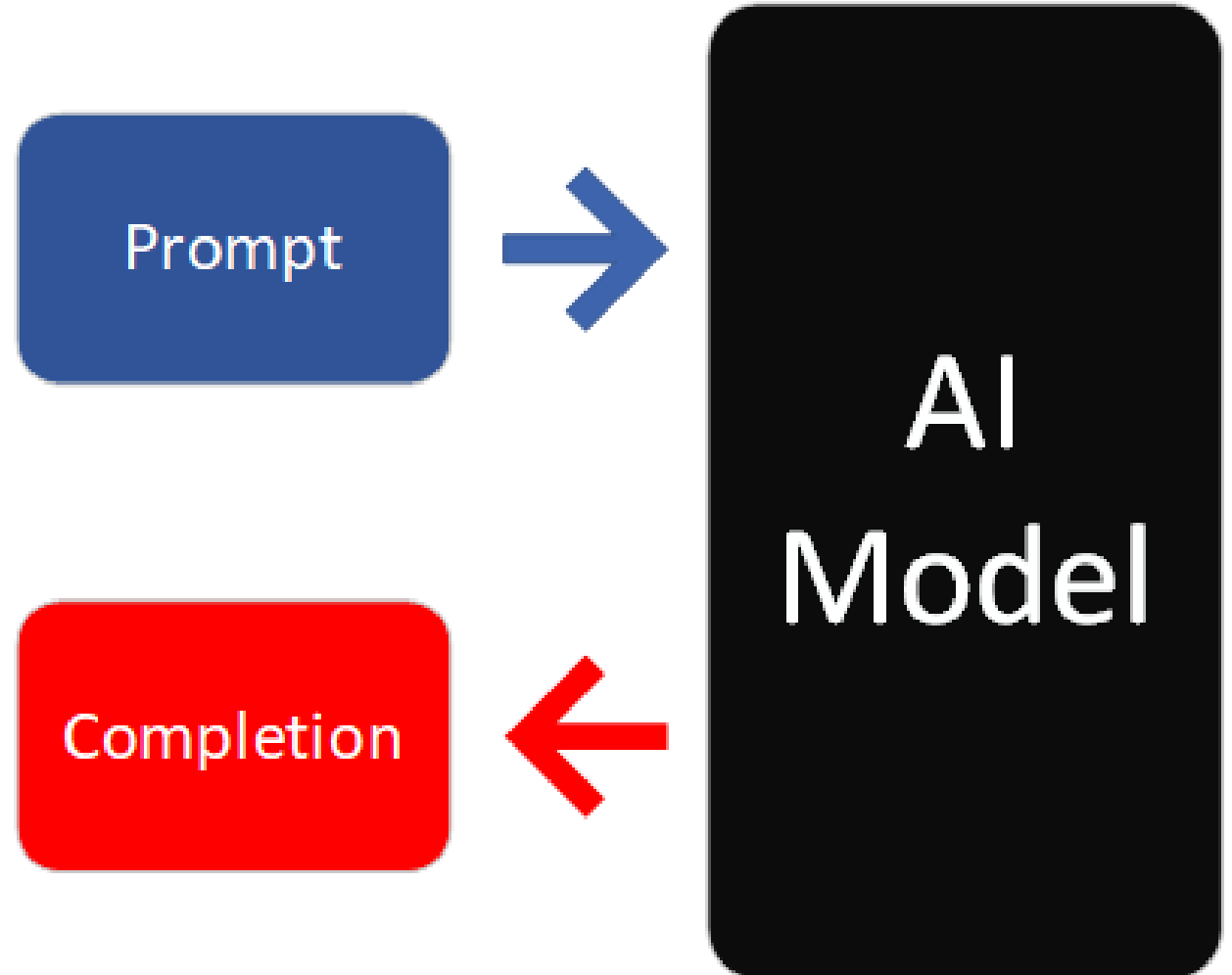
Prompt engineering



Understanding prompts, completions, and tokens

Prompt engineering

Like a person writing an essay, an AI model takes a prompt and continues writing based on the text in the prompt. The new text that the model outputs is called the *completion*. An example task might be to write a Python program to add two numbers. If you write out the task as a Python comment like so:

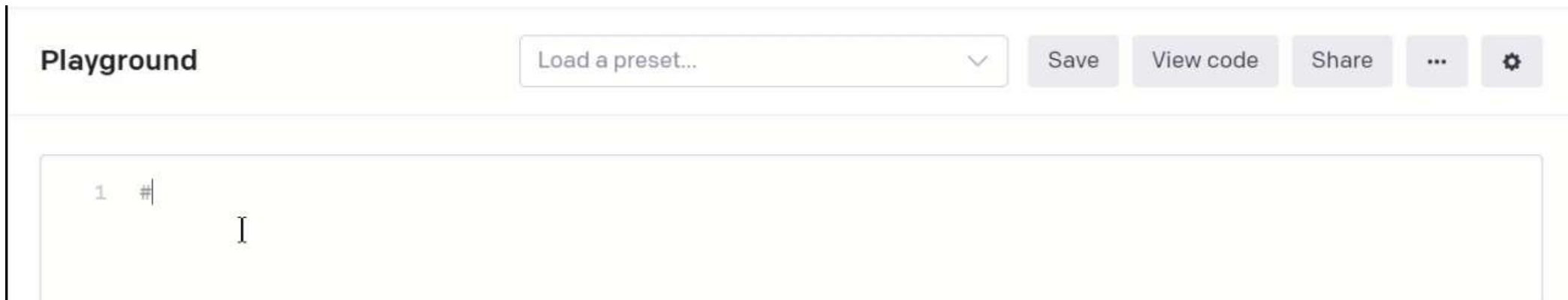


Understanding prompts, completions, and tokens

Write a function that adds two numbers and returns the result.

t

```
def add(a, b):  
    return a + b
```



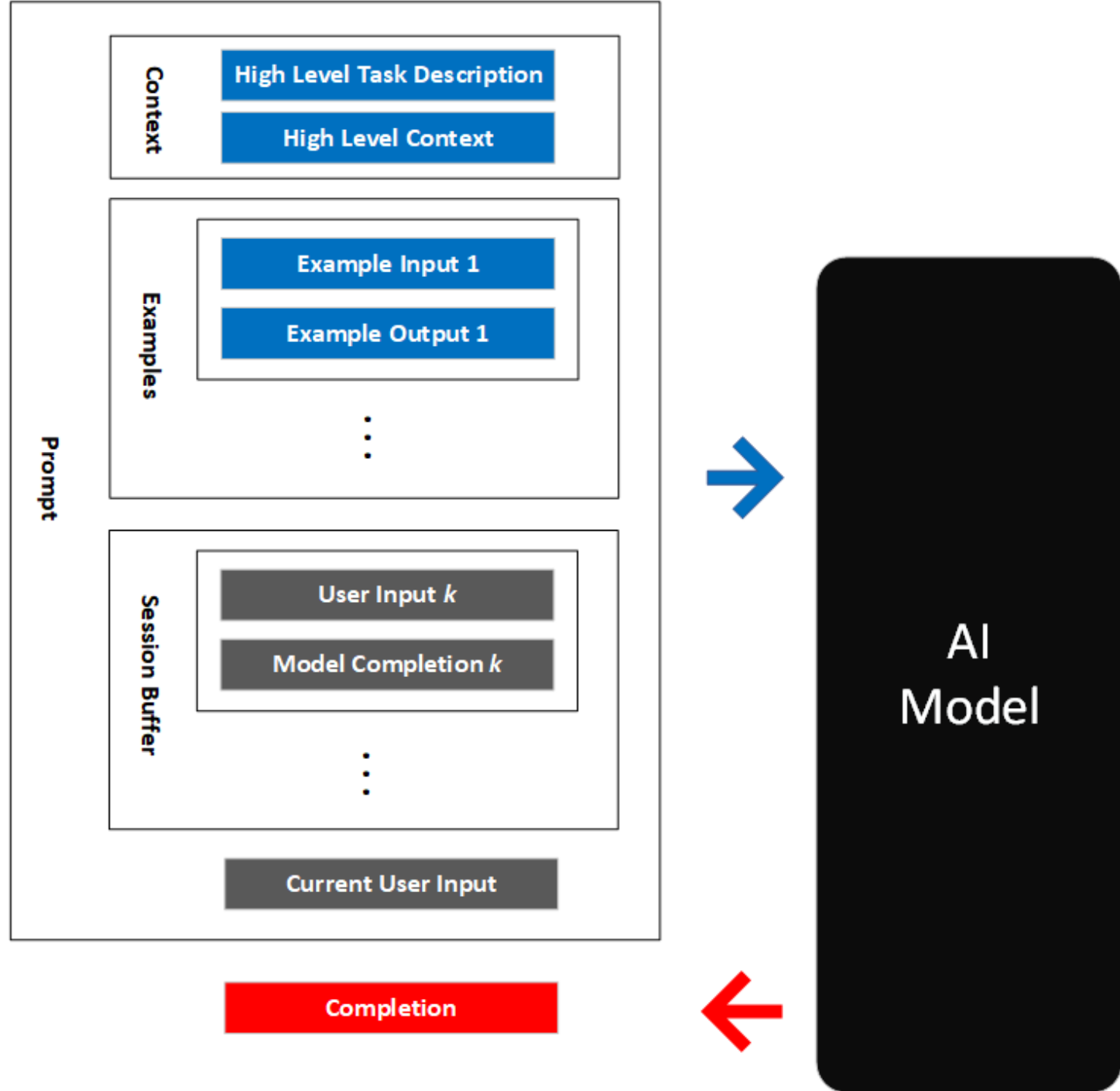
Understanding prompts, completions, and tokens

```
# Load iris data from scikit-learn datasets and plot the training data.
```

```
1 # Load iris data from scikit-learn datasets and plot the training data.
```

```
2
```





API Call

```
#Note: The openai-python library support for Azure OpenAI is in preview.
```

```
import os
```

```
import openai
```

```
openai.api_type = "azure"
```

```
openai.api_base = "https://docs-test-001.openai.azure.com/"
```

```
openai.api_version = "2022-12-01"
```

```
openai.api_key = os.getenv("OPENAI_API_KEY")
```

```
response = openai.Completion.create(
```

```
    engine="gpt-35-turbo",
```

```
    prompt="<|im_start|>system\nYou are an Xbox customer support agent whose primary goal is to help users v
```

```
    temperature=0,
```

```
    max_tokens=100,
```

```
    top_p=0.95,
```

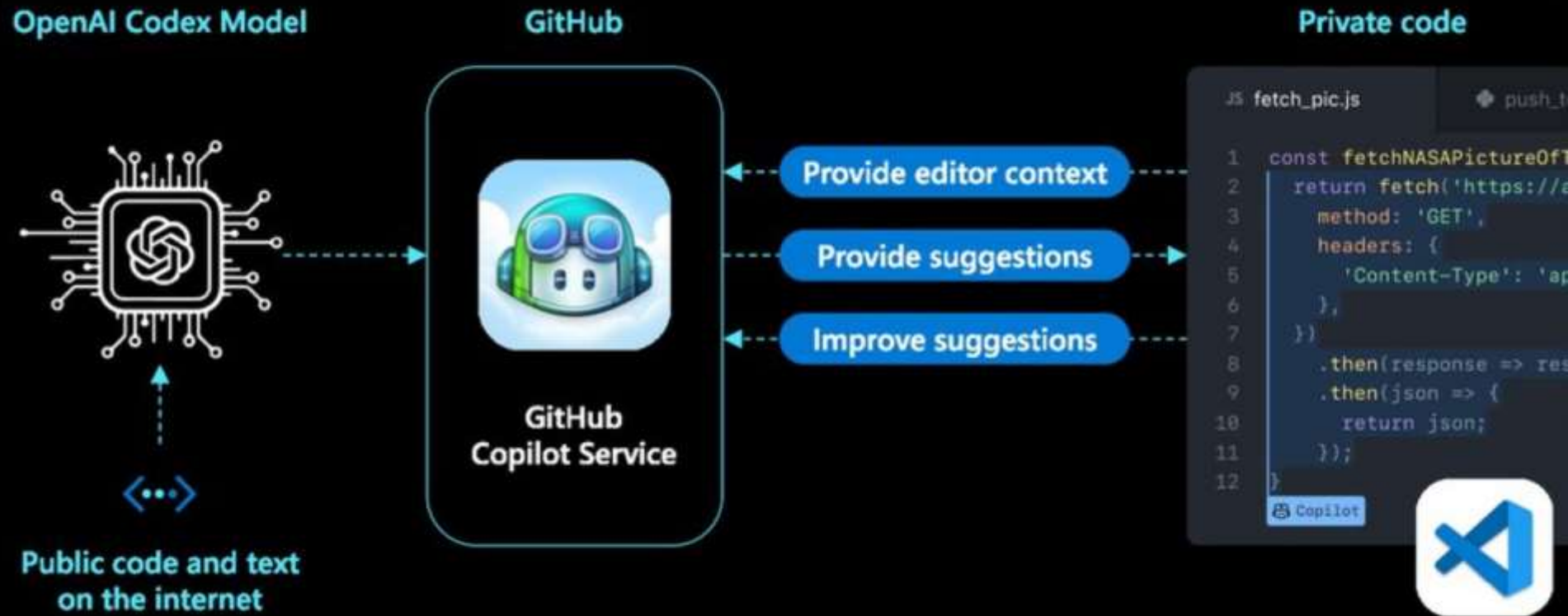
```
    frequency_penalty=0,
```

```
    presence_penalty=0,
```

```
    best_of=1,
```

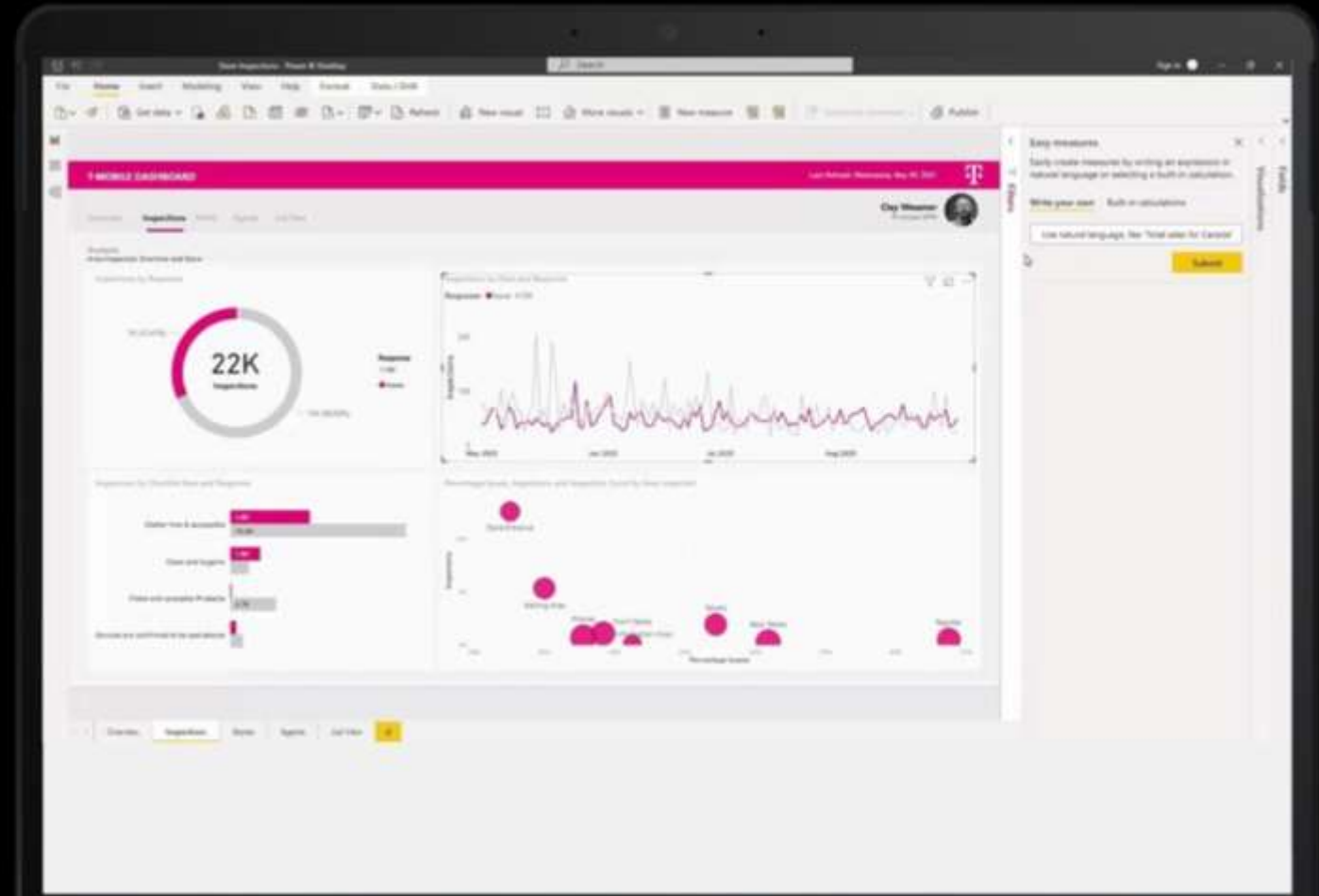
```
    stop=["<|im_end|>"])
```

Usecase #1



Usecase #2

Power BI
Generating DAX
expressions with
natural language



Playground

Cognitive Services | Azure OpenAI Studio

Azure OpenAI Studio > GPT-3 playground

Privacy & cookies

GPT-3 playground

Deployments: text-davinci-002

Examples: Load an example

View code

Start typing here

Parameters

Temperature: 1

Max length (tokens): 100

Stop sequences

Top probabilities: 0.5

Frequency penalty: 0

Presence penalty: 0

Best of: 1

Pre-response text

Post-response text

Learn more

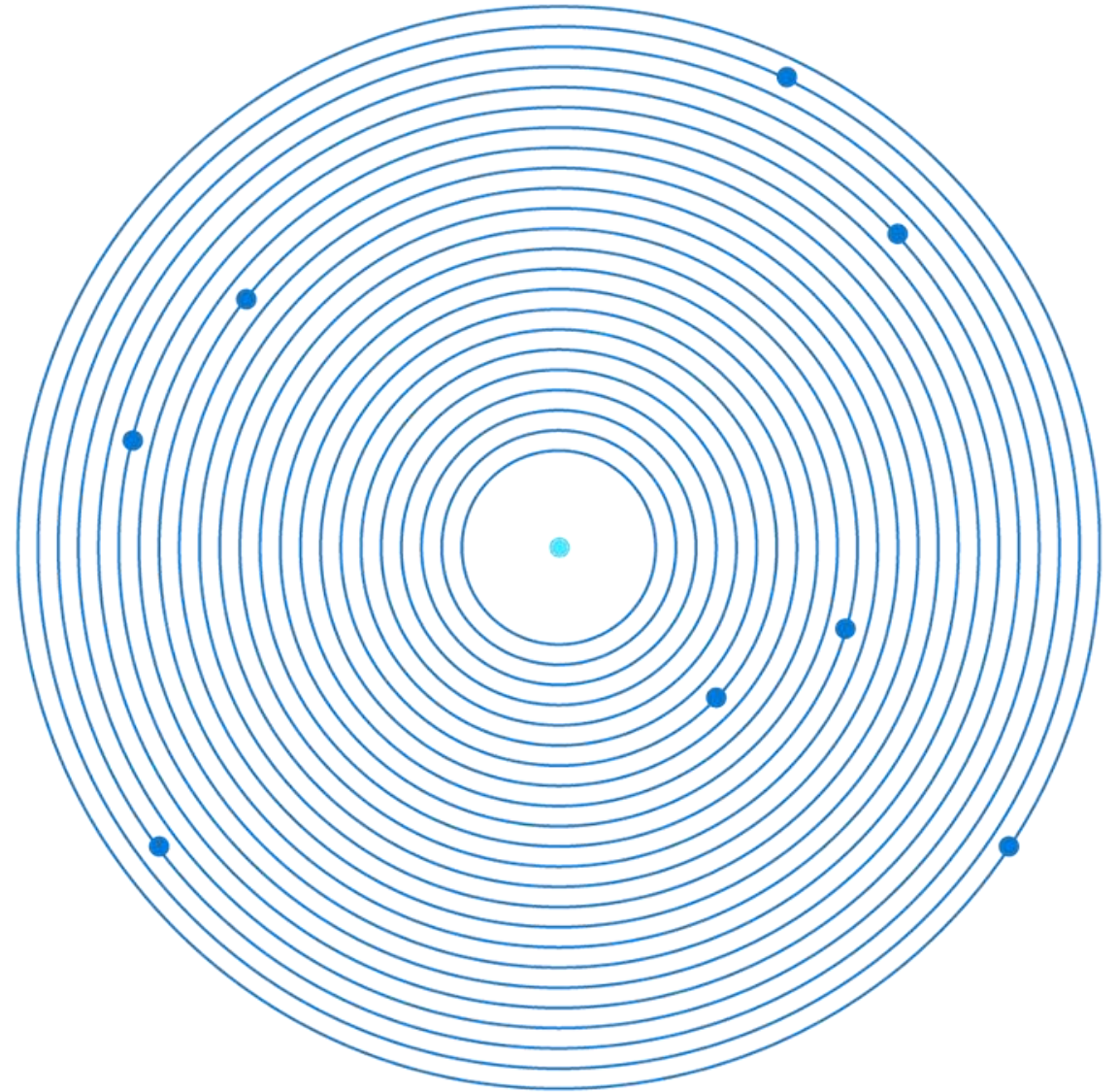
Generate Undo Regenerate Tokens: 0

Q & A

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Thank you 😊

